Project Report Format of Science Exhibition to be filled by Student

Brief about the work undertaken (strictly 600 - 1,200 words) in the following format.

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School: Udaan International School, Begusarai, Bihar

Sub-Theme: Artificial Intelligence (AI) for Better Life

Mentor: Soni Kumari (PGT Science) - Date: 12 November 2026

I. TITLE OF THE PROJECT (Maximum 40 words):

KRISHI-DRISHTI: A Portable Embedded Camera and Edge AI Prototype for Early Foliar Disease Screening in Tomato Crops for Smallholder Farmers

II. NATURE OF WORK (Tick one of the 03 categories):

[ ] Application of basic principles of science and technology

[X] Novel solution/ indigenous design to a problem/ challenge

[ ] Investigation based study

III. RATIONALE (Need for the work):

During the nationwide tomato price spike (>Rs 200/kg), smallholder growers in eastern India suffered over 40% yield losses from foliar blights (Early and Late Blight). In rural districts like Begusarai, marginal farmers lack nearby pathology labs, and weak mobile connectivity makes cloud apps fail in open fields. Anxious growers spray expensive chemical fungicides blindly (~Rs 2,150/acre), escalating debt and harming soil. KRISHI-DRISHTI solves this as an affordable, handheld, offline-first prototype under Rs 1,500 (unit BOM: Rs 1,480) providing instant preliminary foliar screening without internet or recurring fees.

IV. SCIENTIFIC PRINCIPLES AND CONCEPTS USED:

1. Plant Pathology (Biology): Early Blight (Alternaria solani) causes concentric target-board spots. Late Blight (Phytophthora infestans) causes fast-spreading water-soaked lesions in humid air (>85% RH). Early screening enables spot bio-control before fields are lost.

2. Optics & Standardization (Physics): Sun glare and hand-shake mislead computer vision. An 8 cm tubular shroud with a matte acrylic diffuser fixes focal distance and eliminates point-source LED shadows.

3. Dual-Microcontroller Architecture (Electronics): An ESP32-CAM captures 1600x1200 frames via DMA PSRAM to an SD card. An ESP32 DevKit V1 controller runs an SSD1306 OLED (I2C GPIO 21/22) and reads a DHT11 sensor (GPIO 18). We linked the wand to the camera over Wi-Fi at 8.5 dBm TX power, using static IP 192.168.4.2 with DHCP fallback. The connection retries non-blocking every 5 seconds so the NEXT button keeps working even while the wand is still linking up.

4. Quantized Edge AI & Gating (Computer Science): A Float16 MobileNetV3 model (2.01 MB) runs inside a phone browser via WebAssembly over SoftAP (192.168.4.1). Shannon Entropy gating (H > 1.10) flags ambiguous scans, safely withholding chemical advice.

V. MATERIALS AND EQUIPMENT USED (Standardized BOM under Rs 1,500):

Subsystem / Node | Technical Specification | Function in Prototype | Unit Cost
ESP32-CAM Node | Dual-core 240MHz, 4MB PSRAM, OV2640/OV5640 | Image capture, SoftAP web server, SD logger | Rs 540
ESP32 DevKit V1 | Dual-core 240MHz, 30-pin, ESP-NOW link | Master controller, button logic, OLED driver | Rs 420
SSD1306 OLED | 0.96 inch, 128x64 px, I2C (GPIO 21/22) | 4-page status and telemetry carousel | Rs 190
MicroSD Card (8 GB) | Class 10, FAT32 format, 1-bit SD_MMC | Image storage and offline WebAssembly assets | Rs 160
DHT11 Sensor | Temp 0-50 C, Humidity 20-90% (GPIO 18) | Microclimate disease risk tracking | Rs 95
Optical Shroud | 8 cm focal tube, matte acrylic diffuser | Standardizes focal distance and stops glare | Rs 45
Passives & Switches | Tactile switches (GPIO 26/27), 470uF cap | Debounced buttons and power rail protection | Rs 30
Total Bill of Materials | 100% COTS Components | Complete Handheld Screening Prototype | Rs 1,480

VI. STEP-BY-STEP PROCEDURE OF WORK & EXPERIMENTAL METHODOLOGY:

Step 1: Circuit Assembly - Wired OLED (I2C GPIO 21/22) and DHT11 (GPIO 18) to ESP32 controller. Soldered 470uF cap across camera 5V rail to prevent flash brownouts.

Step 2: Optical Shroud Alignment - Fitted 8 cm tubular shroud with matte diffuser over camera to fix focal plane and prevent sunlight glare from washing out leaf textures.

Step 3: Firmware Setup - Programmed both chips in Arduino IDE: camera buffers 1600x1200 frames to SD; controller manages debounced buttons (GPIO 27 SCAN, GPIO 26 NEXT) for OLED display.

Step 4: Edge AI Co-Processor - Loaded Float16 MobileNetV3 and WebAssembly runtime on SD card. Connecting phone to SoftAP (192.168.4.1) runs neural inference 100% offline.

Step 5: Testing & Gating - Benchmarked 60 botanical leaves and audited outdoor images, using Shannon Entropy gating (H > 1.10) to intercept ambiguous foliage and withhold sprays.

1. Variables & Parameters:

Independent: Optical focal shroud (8cm shroud vs. unconstrained) and lighting (direct sun vs. diffused).

Dependent: Write duration (ms), confidence (%), Shannon entropy (H), and diagnostic precision.

Controlled: Resolution (1600x1200 UXGA), JPEG quality (10), voltage (3.3V/5V), and Float16 weights.

2. Scientific Hypothesis: An 8 cm optical shroud and diffuser with Shannon Entropy gating standardizes leaf lighting and enables reliable sub-second foliar disease screening offline, withholding chemical advice when ambiguity exceeds H > 1.10.

3. Safety Precautions:

Electrical: Multimeter verified sensitive 3.3V camera pins never received over-voltage.

Battery: Protected 5V power bank prevented thermal risks during outdoor tests.

Hygiene: Handled diseased leaves with gloves, sealing infected samples in bags.

VII. OBSERVATIONS, RESULTS, AND SCIENTIFIC CONCLUSIONS:

1. Botanical Benchmark (N = 60 Leaves): Under the shroud, the system achieved 88.33% strict accuracy and 90.00% pathogen family agreement. Shannon Entropy gating intercepted 3 ambiguous samples (H > 1.10), lifting accepted clinical accuracy to 92.98%.

2. Outdoor Benchmark & Shroud Necessity: On unconstrained outdoor datasets (PlantDoc), raw accuracy dropped to 33.3% due to direct noon glare and foliage shadows mimicking blight lesions. This proved KRISHI-DRISHTI's physical shroud and entropy gating are essential: without them, mobile cameras generate false alarms; with them, lighting is controlled and ambiguous scans are safely gated.

3. Sub-Second Latency: Full cycle executed in 420 ms (14 ms packet, 19 ms neural inference, 387 ms PSRAM/SD write) entirely without cellular data.

Limitations:

Preliminary Screening: Provides decision support, not certified lab cultures. Suspected cases require agronomist review.

Foliar Scope: Inspects leaf surfaces only; cannot detect root rot or vascular wilt.

Operation: Shroud must rest flat against the leaf surface to avoid edge blurring.

Agricultural & Economic Implications: KRISHI-DRISHTI proves smallholder farmers can access reliable AI in rural fields for under Rs 1,500 (unit BOM: Rs 1,480). Avoiding 2-3 redundant chemical sprays saves Rs 4,300-Rs 6,800/acre annually while protecting soil health.

VIII. REFERENCES & CITATIONS:

Agrios, G. N. (2005). Plant Pathology (5th ed.). Academic Press.

ICAR-NCIPM. Integrated Pest Management Package for Tomato. New Delhi.

FAO. (2021). The State of Food and Agriculture. Rome.

Howard, A. G., et al. (2017). MobileNets for Mobile Vision. arXiv:1704.04861.

CBSE. (2026). Guidelines: CBSE Science Exhibition 2026-27 (Cir. Acad-69/2026).

Official Attestation: We certify that this report presents our authentic original project work executed under mentor guidance. Physical signatures, mentor recommendation forms, and institutional stamps are submitted alongside this dossier.

Team ID: T26706431 - Category II

Gautam Kumar & Abhinav Kumar (Class IX) - Udaan International School, Begusarai - 12 November 2026